INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

THEODOLITES, SURVEYING, DIRECTION: SECOND ORDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of night-illuminated direction theodolites with accessories, cases, and tripods.

1.2 Classification. The theodolite shall be one of the following types, as specified (see 6.2):

Type I - 1 second
Type II - 0.002 mil

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC FSC 6675

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-208	- Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces.
V-T-276	- Thread, Cotton.
W-L-111	- Lamp, Electric Incandescent, Miniature, Tungsten-Filament.
QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
TT-F-325	- Filler, Engraving, Stamped Marking.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and After Treating Processes for Cotton Cloths.
NNN-P-40	- Paper, Lens.

MILITARY

MIL-B-18	- Battery, Dry.
MIL-B-18/9	- Battery, Dry, BA-30.
MIL-V-173	- Varnish, Moisture and Fungus Resistant, for Treatment of Communications, Electronic and Associated Equipment.
MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-H-639	- Hooks (for Belts, Straps and Equipage Items).
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-3530	- Thread and Twine; Mildew Resistant or Water Repellent Treated.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.
MIL-T-14091	- Tripod, Surveying: Extension-Leg-Type.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-43566	- Tape, Textile, Cotton, General Purpose, Natural or in Colors.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standard for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MS15611	- Lamp, Incandescent, G-3-1/2, Miniature Screw, Clear.
MS35335	- Washer, Lock, Flat-External Tooth (In./MM).
MS51957	- Screw, Machine-Pan Head, Cross-Recessed, Corrosion Resisting Steel, UNC-2A.
MS51959	- Screw, Machine-Flat Countersunk Head, 820, Cross-Recessed, Corrosion Resisting Steel, UNC-2A.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 1911-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the specification.

NATIONAL BUREAU OF STANDARDS (NBS)

Special Publication 374	- Method for Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.)

2.2 Non-government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rules for the Use of Volatile Organic Compounds (VOC).

(Applications for copies should be addressed to the South Coast Air Quality Management District, Attn: Public Affairs Department, 9150 Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The theodolite shall be an optical-reading direction-type instrument capable of second-order accuracy (see 6.6). The theodolite, including the alidade and tribrach, shall weigh not more than 17 pounds 10 ounces, or 8.0 kilogram (kg). The theodolite shall be furnished with a carrying case, a transport case, a plastic theodolite cover, and when specified, a battery and lighting accessory case, tools and accessories, and a tripod.

3.2 First article (first-produced theodolite). Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The theodolite shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the theodolite may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the theodolite may be newly fabricated from recovered materials to the maximum extent practicable, provided the theodolite produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the theodolite.

3.4 Environmental conditions.

3.4.1 Operating temperature. The theodolite, tripod, cases, and accessories shall operate as specified herein and shall not be damaged when subjected to ambient temperatures from +125 °F (51.7 °C) to -40 °F (-40 °C).

3.4.2 Storage temperature. The theodolite, cases, and accessories shall not be damaged by storage under the following conditions:

Air Temperature:

High - +155 °F (68.3 °C) Time period - 4 hours.
Low - -50 °F (-45.6 °C) Time period - 6 hours. No solar radiation.

3.4.3 Solar radiation. The theodolite shall operate as specified herein and shall not be damaged by full impact of solar radiation of 355 British thermal units per square foot per hour (Btu/ft^2/hr) at an air temperature of 125 °F (51.7 °C) for 4 hours.

3.4.4 Rainfall. The theodolite, carrying case, and transport case shall show no signs of water penetration when subjected to rainfall at a rate of 3 inches, ±1 inch, per hour, as follows:

Theodolite	- 5 minutes
Carrying case	- 1 hour
Transport case	- 1 hour

3.4.5 Sand and dust. When the theodolite and carrying case are subjected to blowing sand and dust, the theodolite shall not be damaged, shall not show sand and dust penetration in the telescope, in the optical system, electronics package, or on bearings, and shall not show lens abrasion. The carrying case shall not show sand and dust penetration.

3.4.6 Vibration. The theodolite and carrying case shall not be damaged by vibration when tested as specified in 4.6.2.18.

3.4.7 Shock. The theodolite and cases shall not be damaged by shock when tested as specified in 4.6.2.19.

y of Mil products, associated equipment and personnel. (3) Security

3.4.8 Humidity. The item shall not be damaged and shall operate within the accuracy specified herein when subjected to relative humidity of at least 95 percent at air temperature varying between -25 °F (-32 °C) and +110 °F (+43 °C).

3.5 Alidade assembly. The alidade assembly shall include the alidade proper and the lower motion. The alidade proper shall include the vertical circle, the micrometer, the plate and altitude levels on the self-indexing device, the telescope, the horizontal axis, and wiring for electric illumination. The lower motion shall include the horizontal circle, the vertical axis, provision for electric illumination, and a means for attaching to the tribrach. The alidade shall display an erect image.

3.5.1 Telescope. The telescope shall consist of the optical system, sights or a collimator for aiming, and a retractable or removable sunshade. The overall length of the telescope shall be not greater than 180 millimeters (mm) when the diopter scale is set to zero. The telescope shall be mounted so that pointings may be made to the zenith by use of the elbow eyepiece. A means for illuminating the reticle shall be internally mounted in the telescope, and a means for controlling the illumination shall be provided. The centering of the telescope over the vertical axis shall be such that the combined error and the change in the line of collimation resulting from the travel of the negative lens over the focusing range shall not change the line of collimation by more than 10 seconds of arc for type I and 0.05 mil for type II instruments over a focusing range from 15 feet (4.6 meters) to infinity. The telescope shall balance when focused at infinity with the sunshade in the extended position.

3.5.1.1 Optical system. The optical properties shall conform to the following requirement:

Magnification, telescope	- 28 diameters, +4 or -1 diameter
Effective aperture of objective, (min)	- 40 mm
Field of view (min)	- 1 degree 20 minutes, or 24 mils
Focus, minimum, not greater than	- 2.2 meters, or 7 feet
Resolution	- 4 seconds or better of arc

The principal axis of the lens system shall be in alignment and coincident with the collimation of the telescope in adjustment. The field of view shall be substantially flat from the center to a zone outside the stadia lines equal to 1/4 of the stadia interval, and shall show no marked curvature from there to the edge of the field. Except for the registered face of the reticle, the internal lens-air surfaces of the telescope shall be coated with a reflection-reducing film in accordance with MIL-C-675.

3.5.1.2 Reticle. Unless otherwise specified (see 3.23.7 and 6.2), the reticle shall consist of a pattern of opaque lines similar to that shown in figure 2 (see 6.5), registered on optical glass. The single vertical

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crossline shall subtend equal angles with the adjacent double lines within ±0.5 second. The width of the lines shall be 2.5 to 4.0 microns. The double vertical lines shall subtend an apparent angle of 40 seconds, +5 seconds or -10 seconds. The radius of the solar circle shall be 15 minutes 30 seconds ±10 seconds. The horizontal and vertical stadia lines shall be so spaced that 1/100 of the distance from a stadia rod at a known distance greater than 100 feet or 30 meters from the instrument shall be subtended with an error in reading not to exceed 0.3 percent. The lines shall subtend 5 to 7 minutes of arc in length and shall be parallel to the corresponding crosslines. The reticle shall be mounted in a metal cell which shall not become distorted during adjustment for collimation. The horizontal and vertical collimation adjustment of the reticle shall be positive and easily accessible. The horizontal and vertical collimation errors of the line of sight shall be not more than 6 seconds (0.03 mil).

3.5.1.3 Telescope eyepiece. The eyepiece shall be so designed that the operator can place his eye in a position coincident with the exit pupil. The eyepiece shall be equipped with a diopter scale marked from zero in plus and minus units, and shall have a index mark for referencing the setting. The telescope eyepiece shall be designed for placing of the elbow eyepiece (see 3.5.1.4) and interchange with the autocollimation eyepiece (see 3.5.1.6).

3.5.1.4 Elbow eyepiece. Unless otherwise specified (see 6.2), an elbow eyepiece shall be furnished. The eyepiece shall enable an operator to observe vertical angles to the zenith and to observe a ground point or azimuth mark with the telescope in either the direct or reverse position. After transiting the telescope, the elbow eyepiece shall be movable to the observing position without disturbing the levelment of the theodolite. Magnification of the telescope shall not be decreased by use of the elbow eyepiece.

3.5.1.5 Solar filter. Unless otherwise specified (see 6.2), a filter for solar observation shall be provided and shall fit both the telescope eyepiece and the elbow eyepiece. The solar filter shall contain a black or Nichrome-coated (or equal) neutral filter with a density range of between 5.0 and 6.0. If a Nichrome-coated (or equal) filter is furnished, the coated surface shall be toward the eyepiece lens.

3.5.1.6 Autocollimation eyepiece. When specified (see 6.2), an autocollimation eyepiece shall be furnished. The eyepiece shall be interchangeable with the normal eyepiece. The magnification of the telescope with the autocollimation eyepiece installed shall be not less than 18 diameters.

3.5.2 Standards. The standards may be adjustable for height or may be factory adjusted and set permanently. The horizontal axis of the telescope shall be so nearly perpendicular to the vertical axis of the theodolite that the error due to the difference in the height of the standards shall be not more than 12 seconds or 0.06 mil of arc.

3.5.3 Optical plummet. The optical plummet may be integral with the alidade or may be attached to the tribrach and shall be adjustable. The reticle pattern shall have 4 sectors of 90 degrees; and, when specified (see 6.2), the opposite sectors shall be frosted as shown on figure 3. The optical system shall produce a clear image throughout the operating range from tripod height to not less than 110 feet (33.5 meters). The line of sight of the

3.6.6 Elbow eyepiece. When the reading microscope is attached to the telescope and cannot be read in every position of the telescope, an elbow eyepiece for the microscope shall be furnished, unless otherwise specified (see 6.2). The eyepiece shall give an erect image of the reading scales. The use of the elbow eyepiece shall not change the magnification of the microscope.

3.7 Levels. The instrument shall be equipped with a plate level, circular level, altitude level or a self indexing device, where applicable.

3.7.1 Plate level vial. The plate-level vial shall be graduated at 2 mm intervals and shall have a sensitivity of 20 seconds, ±4 seconds. Graduations shall be filled with a paste conforming to TT-F-325, type II. Graduations shall be symmetrical about the exposed portion and shall extend far enough to include ends of the bubble at the operating temperatures specified herein. The plate level shall be mounted as close as practicable to the vertical axis and shall be protected. A device for adjusting the level shall be provided.

3.7.2 Circular level. The instrument shall be equipped with a circular level. The circular level shall have a sensitivity of 10 minutes, ±3 minutes for each 2 mm. It shall be mounted on the alidade or on the tribrach.

3.7.3 Vertical circle indexing. The indexing of the vertical circle shall be controlled by a self-indexing device, except that, when specified (see 6.2), an altitude level shall be furnished. When the vertical circle in the direct position reads 90 degrees or 1600 mils, the telescope shall be horizontal.

3.7.3.1 Self-indexing device. The self-indexing device shall be damped so that oscillation is stopped in not more than 5 seconds of time. The device shall be accurate within 10 seconds of arc or 0.05 mil, as applicable.

3.7.3.2 Altitude level. The altitude level for indexing the vertical circle shall have a sensitivity of 30 seconds, ±5 seconds, from each 2 mm, and shall be ungraduated. The level shall be read by the coincidence method through a train of prisms and a rotatable viewing glass on the vertical circle standard. The image of the coincident bubble ends shall appear in the viewing glass at any operating temperature specified herein. The level shall be adjustable. A tangent screw shall be provided for bringing the bubble ends into coincidence in the viewing device

3.8 Bearings. The rotating alidade shall be supported on a bearing. The telescope shall be supported on bearings, and no lateral motion shall exist between the telescope and standards.

3.9 Tribrach. The tribrach shall consist of the leveling screws, base plate, and a device for locking the alidade assembly. The base plate shall have at its center a 5/8-inch nut of the unified coarse thread series (5/8-1 UNC-2B), class 2B, 11 threads per inch, as defined in FED-STD-H28. The

leveling screws shall permit leveling of the theodolite when the base plate is 6 degrees out of level.

3.9.1 Maintenance of centering. The design of the tribrach shall be such that removing and returning the alidade shall cause a change in the centering of the alidade not to exceed 0.002 inch (0.05 mm).

3.9.2 Interchangeability. Tribrachs shall be interchangeable with all alidades procured under this specification from the same manufacturer.

3.10 Fastening screws. Size, type, form, and class of fit for fastening screws shall be selected by the contractor as best suited for their intended use. There shall be no exposed threads on the theodolite except tapped holes.

3.11 Operating screws. Operating screws shall have heads designed for recognition in darkness. The heads of the altitude-level tangent screw and the vertical-motion tangent screw shall be distinctly different in shape. Leveling screws and tangent screws shall have a take-up for wear adjustment and shall be removable for cleaning.

3.12 Illumination. The theodolite shall be illuminated by natural and electrical means through not more than two openings in the shell.

3.12.1 Natural illumination. Natural illumination shall be from a second surface mirror or mirrors which shall be adjustable to any orientation. Moving or adjusting the mirror shall not cause a change of more than 1 second or 0.005 mil in the micrometer reading.

3.12.2 Electric illumination. Power for electric illumination shall be obtained from the batteries. When the theodolite is operated under blackout conditions, there shall be no light visible beyond 100 feet except that in the telescope field of view.

3.12.2.1 Wiring. If wiring is used for the lighting equipment, sufficient slack shall be allowed for field repair or broken connections. The slack shall not in any way impede or interfere with the operation of the theodolite. The wire shall be extra-flexible and shall not show evidence of damage, nor crack at -50 °F (-45.5 °C) when wrapped around a 3.8-inch diameter (9.7 cm) mandrel.

3.12.2.2 Hand lamp. When specified (see 6.2) a hand lamp shall be provided and shall connect to a receptacle on the battery case. The hand lamp shall have a switch and a connecting cord not less than 5 feet (1.5 meters) long. The socket of the hand lamp shall be threaded with the miniature lamp holder thread as defined in FED-STD-H28. The cord shall meet the requirements of 3.12.2.1.

3.12.2.3 Bulbs. If bulbs are used in the lighting attachment and in the hand lamp, the bulbs shall conform to W-L-111.

3.12.2.4 Battery and lighting accessory case. When specified (see 6.2) a battery and lighting accessory case of metal, glass fiber reinforced plastic laminate, or a high-impact plastic that will withstand dropping under the low temperature specified in 3.4.1 without breakage or damage shall be furnished. The case shall contain space for four batteries conforming to MIL-B-18 and MIL-B-18/9. The batteries shall not be furnished. The batteries shall be connected in a series-parallel arrangement to provide 3 volts. The case shall be provided with two receptacles. One receptacle shall be equipped with a rheostat with a marked "OFF" position. The other receptacle shall provide full voltage. The interior of the case shall provide space for the hand lamp, the connector cords if applicable, four replacement lamps, and the four batteries. The case shall have a provision for mounting on the tripod.

3.12.2.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.12.2.5.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.2.5.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with the rules associated with volatile organic compounds (VOC) as established by local air quality rulings and South Coast Air Quality Management District rulings (see 6.2).

3.13 Carrying case. A carrying case of metal, high-impact molded plastic, or glass fiber reinforced plastic laminate shall protect the theodolite against damage. Each case shall contain a desiccant adequate to the size of the container and provision for storage of the adjusting pins and jewelers screwdriver. The glass fiber reinforced plastic laminate case shall have reinforced corners. The theodolite shall be fastened within the case to prevent the theodolite from falling out when the case is opened. The case shall be provided with heavy-duty catches for closing. If the glass fiber reinforced plastic laminate case contains a filler of plastic foam and skin or

other material, the filler shall be removable for cleaning. The weight of the carrying case shall not exceed 13 pounds (5.9 kg).

3.14 Transport case. A reusable transport case which will house and protect the theodolite in its carrying case against rainfall, shock, and vibration during field transport, shipment, and storage shall be furnished. The transport case shall not be damaged, shall protect the theodolite in its carrying case from damage, and shall not allow water to enter the case when subjected to the shock and rainfall tests specified in 4.6.2.14.3 and 4.6.2.19. The case shall be made of molded thermoplastic resin or glass fiber reinforced plastic laminate, and shall weigh not more than 15 pounds (6.8 kg). The case shall have removable inserts contoured to fit snugly around the carrying case. The case shall have carrying handle(s) and lock(s) or clasp(s). Glass fiber reinforced plastic laminate cases shall have reinforced corners. All hardware shall be of corrosion-resistant metal or shall be of metal treated to resist corrosion.

3.15 Theodolite cover. A lightweight plastic flexible waterproof cover shall be furnished to protect the theodolite from rain and dust. The cover shall withstand the environmental conditions specified herein without damage.

3.16 Winterization. Control knobs and leveling screws shall have plastic or plastic coated heads.

3.16.1 Case. The case shall be of cotton duck and shall contain all the components of the winterization kit. The case shall have a slide fastener, a belt loop, and a double belt hook conforming to MIL-H-639, type I, class 1. The cotton duck used shall conform to CCC-C-419, type I, no. 12, hard-texture and rendered fungus resistant and water repellent in accordance with CCC-D-950, type I, class B, inhibitor (f), and dyed olive drab 7. Webbing used shall conform to MIL-W-530, type IIa, class 4, dyed olive drab 7. Tape shall conform to MIL-T-43566, type I, class 4. Thread used in fabricating the case shall conform to V-T-276, type III B, ticket no. 10/4, dyed olive drab 7, and rendered fungus resistant in accordance with MIL-T-3530, type I, class 1.

3.17 Knapsack. When specified (see 6.2), a knapsack shall be furnished for backpacking the theodolite in its carrying case. Shoulder straps shall be cushioned or padded. The knapsack shall be fabricated from cotton duck conforming to CCC-C-419, type I, no. 12, hard texture, and rendered fungus resistant and water repellent in accordance with CCC-D-950, type I, class B, inhibitor f. Webbing shall conform to MIL-W-530, type IIa, class 4 and be made water repellent and fungus resistant. Tape shall conform to MIL-T-43566, type I, class 4. Thread used in the fabrication of the knapsack shall conform to V-T-276, type IIIb, ticket no. 10/4 and rendered fungus resistant in accordance with MIL-T-3530, type I, class 1. The knapsack shall be fitted with packets for carrying accessories and shall be large enough to allow for shrinkage of the canvas. A cap shall fit snugly over the top of the knapsack to prevent penetration of rain. The color of the knapsack shall be as specified (see 6.2).

3.18 Lubrication. The theodolite shall be lubricated prior to assembly with grease conforming to MIL-G-23827. The grease shall permit the theodolite to operate throughout the specified temperature ranges.

3.19 Identification marking. Each theodolite shall be identified in accordance with MIL-STD-130. The marking of the theodolite shall be placed on one of the standards. Markings shall include: The letters "US", the contractor's name or trademark, the serial number of the theodolite, the type marking "DEG" or "MIL", as applicable, the NSN assigned to the theodolite, the contract number under which the theodolite was procured, and the year of manufacture. The carrying case shall be marked with the letters "US", the type 1 designation, as applicable and the serial number of the theodolite. The markings just specified for the carrying case shall be painted on the transport case. When specified, canvas case and the knapsack shall be identified by stenciling using ink conforming to A-A-208.

3.20 Treatment and painting. The portions of the theodolite, cases, and accessories normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color shall be as specified (see 6.2). The interior of the telescope and the optical train shall be matte black.

3.21 Tools and accessories. When specified (see 6.2) the following items shall be furnished with each theodolite:

Item	Quantity
Brush, lens, camel's hair	1
Chamois skin, 8 x 8 inches (20 x 20 cm)	1
Container, lubricant, with not less than 1/4 ounce (7.4 ml) of grease conforming to MIL-G-23827 or equal	1
Cover, waterproof (see 3.15)	1
Elbow eyepiece, telescope (see 3.5.1.4) (unless the eyepiece is not required)	1
Elbow eyepiece, microscope (see 3.6.6) (unless the eyepiece is not required)	1
Filter, solar (see 3.5.1.5) (unless the filter is not required)	1
Lighting attachment(s)	1 set (unless contained in the battery and lighting accessory case (see 3.12.2.4)
Pins, adjusting	2
Screwdriver, flat tip	1
Lens tissue, NNN-P-40 or equal	1 pkg.
Orangewood sticks (commercial grade)	1 pkg.
Manual, instruction for use of theodolite, with protective cover (contractor's standard)	1

The contractor may add items to this list as required for field maintenance and adjustments. The tools and accessories shall be contained in a cotton duck case. The case shall be similar in construction to the case specified in 3.16.1. and shall contain all the tools and accessories.

3.22 Type I theodolites.

3.22.1 Circles. Horizontal and vertical circles shall be graduated at 10- or 20-minute intervals, with every degree graduation numbered. The numbering shall be from zero to 359. The numbers shall be symmetrically oriented with respect to the graduations. Graduations and numbers shall be uniform, legible, and permanent.

3.22.1.1 Horizontal circle. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 2 seconds of arc.

3.22.1.1.1 Circle drive. Settings shall be made within ±5 seconds of arc.

3.22.1.2 Vertical circle. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 3 seconds of arc. The vertical circle shall be mounted so that a zenith distance of 90 degrees is read when the telescope is horizontal in the direct position.

3.22.1.3 Micrometer. The graduation interval of the micrometer shall be 1 second. The difference between the mean readings at the lower end of the scale and the mean readings at the upper end of the scale shall be the span of the micrometer, ±1.0 second, for both the horizontal and vertical circles. The run of the micrometer shall be adjusted so that the span is traversed on the scale between adjacent coincidences or settings of the images of both the horizontal and vertical circles. Tolerance for the run of the micrometer shall be not more than ±1 second for the horizontal circle and ±2 seconds for the vertical circle.

3.22.2 Operating screws. The horizontal-motion tangent screw and the vertical slow-motion screw shall rotate the telescope not less than 7 minutes nor more than 20 minutes of arc per revolution. The horizontal and vertical motion tangent screws shall be long enough to produce at least 3 degrees travel.

3.22.3 Tripod. When specified (see 6.2), a tripod conforming to MIL-T-14091 shall be furnished with each type I theodolite.

3.23 Type II theodolites.

3.23.1 Circles. Horizontal and vertical circles shall be graduated at 2-mil intervals, with every 10-mil graduation numbered with 3 figures. The numbering shall be from 000 to 639. The numbers shall be symmetrically

oriented with respect to the graduations. Graduations and numbers shall be uniform, legible, and permanent.

3.23.1.1 Horizontal circle. The circle shall be graduated so that no error in spacing of the graduations either individual or cumulative, shall exceed 0.010 mil.

3.23.1.1.1 Circle drive. Settings shall be made within ±0.025 mil.

3.23.1.2 Vertical circle. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 0.015 mil. The vertical circle shall be mounted so that a zenith distance of 1600 mils is read when the telescope is horizontal in the direct position.

3.23.1.3 Micrometer. The graduation interval of the micrometer shall be 0.002 mil. The difference between the mean readings at the lower end of the scale and the mean readings at the upper end of the scale shall be the span of the micrometer, ±0.005 mil, for both the horizontal and vertical circles. The run of the micrometer shall be adjusted so that the span is traversed on the scale between adjacent coincidences or settings of the images of both the horizontal and vertical circles. Tolerance for the run of the micrometer shall be not more than ±0.005 mil for the horizontal circle and ±0.001 mil for the vertical circle.

3.23.2 Operating screws. The horizontal-motion tangent screw and the vertical slow-motion screw shall rotate the telescope not less than 2 mils nor more than 6 mils per revolution. The horizontal and vertical motion tangent screws shall be long enough to produce at least 53.3 mils of travel.

3.23.3 Illuminated optical plummet. When specified (see 6.2), the optical plummet shall be illuminated by an eyepiece adapter conforming to figure 4.

3.23.4 Clamp assembly. When specified (see 6.2), a clamp assembly shall be installed on the changeover (circle selector) knob.

3.23.5 Collimation light assembly. When specified (see 6.2), a collimation light assembly conforming to figure 5 shall be installed in the telescope of the theodolite.

3.23.6 Tripod. When specified (see 6.2), a tripod conforming to MIL-T-14091 shall be furnished with each type II theodolite.

3.23.7 Reticle. The reticle for type II theodolite shall be the pattern shown in figure 2 (see 6.5), and as specified in 3.5.1.2.

3.24 Workmanship. All threads requiring movement, sliding and rotatable parts, shall move without binding, backlash, or end play. Clamping actions shall be positive and, when unclamped, shall not cause drag. Focusing components or assembles shall retain their focus when finger pressure is released and the telescope shall not change focus when transited. No lateral

motion shall exist between the telescope and standards. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between the glass and coated film, or cloudiness. Optics shall show sharpness of focus and distinct images to provide favorable definition. Metal surfacings on the optical components for all prisms and lenses shall not shift. There shall be no flickering of the illumination system when the vertical or horizontal motions are rotated. Liquid within the level vials shall be clear, and the vials shall be tight in the housings. Components and assemblies shall show no evidence of forced attachment. Bearing surfaces shall show no scratches or surface blemishes. Mirrors shall not be tarnished, and backing shall be uniform and unspotted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests), specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Prior to assembly (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection of packaging (see 4.7).

4.3 Prior to assembly inspection.

4.3.1 Tests. Prior to assembly, the horizontal and vertical circles shall be tested for accuracy, the level vials shall be tested for sensitivity and the reticles shall be measured for reticle line weight as specified in

4.6.2.2. Failure of the applicable test shall be cause for rejection of the applicable item.

4.4 First article inspection.

4.4.1 Examination. The theodolite, carrying case, transport case, battery case, and accessories shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. The theodolite, carrying case, transport case, battery case, and accessories shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.5. Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa.

4.5.2 Examination. Presence of one or more defects shall be cause for rejection of the lot. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1.

4.5.3 Tests.

4.5.3.1 Individual. Each theodolite shall be tested as specified in 4.6.2.3 and 4.6.2.16. Failure of any test shall be cause for rejection of the theodolite.

4.5.3.2 Samples. Sample selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 5 of table I. Failure of any test shall be cause for rejection of the lot.

4.5.3.3 Humidity. Place the theodolite with power off, mounted in the operating position with all accessories, in a humidity chamber conforming to MIL-STD-810, method 507.2 procedure II. The theodolite shall be inspected and tested as specified in 4.6.2.16. Evidence of damage shall constitute failure of this test.

4.6 Inspection procedure.

4.6.1 Examination. The theodolite, carrying case, transport case, battery case, and accessories shall be examined as specified herein for the following defects:

Major

101. Weight of theodolite and cases not as specified (see 3.1 and 3.13).
102. Materials not as specified (see 3.3).

103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the theodolite (see 3.3.4).
107. Alidade assembly not as specified (see 3.5).
108. Telescope will not transit and balance as specified (see 3.5.1).
109. Coated optics not as specified (see 3.5.1.1).
110. Optical system does not produce a sharp image throughout the entire focusing range and is not free from internal reflection, spherical and chromatic aberration, coma, astigmatism, or distortion (see 3.5.1.1).
111. Reticle pattern not as specified (see 3.5.1.2).
112. Reticle lines not clean-cut, appear fuzzy, or do not appear opaque (see 3.5.1.2).
113. Eyepieces not as specified (3.5.1.3 and 3.5.1.4).
114. Elbow eyepieces not furnished or not as specified (see 3.5.1.4 and 3.6.6).
115. Solar filter not as specified (see 3.5.1.5).
116. Graduations on circles and micrometer scale not clear and clean-cut or appear fuzzy (see 3.6 and 3.6.3).
117. Level graduation marking not as specified (see 3.7).
118. Tribrach not as specified (see 3.9).
119. Operating screws not as specified (see 3.11, 3.22.2 and 3.23.2).
120. Illumination system not as specified (see 3.12).
121. Battery and lighting accessory case not as specified (see 3.12.2.4).
122. Fungus and moisture treatment of canvas items not as specified (see 3.12.2.5).
123. Cases not as specified (see 3.13 and 3.14).
124. Theodolite cover not as specified (see 3.15).
125. Knapsack not as specified (see 3.17).
126. Lubrication not as specified (see 3.18).
127. Tools and accessories not as specified (see 3.21).
128. Circles and micrometer scale not graduated as specified (see 3.22.1, 3.22.1.3, 3.23.1 and 3.23.1.3).
129. Workmanship not as specified (see 3.24).

Minor

201. Identification marking not as specified (see 3.19).
202. Treatment and painting not as specified (see 3.20).

4.6.2 Tests. The theodolite, carrying case, transport case, battery case, and accessories shall be tested as specified herein in accordance with the schedule established in table I. After contractor adjustment of the

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theodolite (see 4.6.2.1), readjustment may only be made at the conclusion of the shock, vibration, and extreme environmental tests.

TABLE I. Test schedule.

Test	Paragraph	First Article	Requirement Paragraph	Quality Conformance
1	2	3	4	5
Prior to assembly:				
Circle	4.6.2.2.1	X	3.6	X
Level, sensitivity	4.6.2.2.2	X	3.7.2	X
Reticle Line Weight	4.6.2.2.3	X	3.5.1.2	X
Post assembly:				
Examination	4.6.1	X	3.1 - 3.24	X
Micrometer	4.6.2.4	X	3.6.3	X
Focus-collimation	4.6.2.5.1	X	3.5.1	X
Height of standards	4.6.2.9	X	3.5.2	X
Mirror adjustment	4.6.2.10	X	3.6.3	-
Levelment	4.6.2.13.2	X	3.9	-
Theodolite system	4.6.2.3	X	3.22 & 3.23	-
Sand and dust:				
Theodolite	4.6.2.15.1	X	3.4.5	-
Carrying case	4.6.2.15.2	X	3.4.5	-
Operational	4.6.2.16	X	3.4 - 3.24	-
Low and high temperature:				
Low-temperature operation	4.6.2.17.2	X	3.4.1	-
Operational	4.6.2.16	X	3.4 - 3.24	-
Low-temperature storage	4.6.2.17.3	X	3.4.2	-
Operational	4.6.2.16	X	3.4 - 3.24	-
High-temperature operation	4.6.2.17.4	X	3.4.1	-
Solar radiation	4.6.2.17.6	X	3.4.3	-
Operational	4.6.2.16	X	3.4 - 3.24	-
High-temperature storage	4.6.2.17.5	X	3.4.2	-
Temperature test of wiring	4.6.2.17.7	X	3.12.2.1	-
Operational	4.6.2.16	X	3.4 - 3.24	-
Rainfall:				
Theodolite	4.6.2.14.1	X	3.4.4	-
Carrying case	4.6.2.14.2	X	3.4.4	-
Transport case	4.6.2.14.3	X	3.4.4	-
Vibration:	4.6.2.18	X	3.4.6	X
Operational	4.6.2.16	X	3.4 - 3.24	-
Shock:	4.6.2.19	X	3.4.7	-
Shock test of transport case	4.6.2.19.1	X	3.4.7	-
Operational	4.6.2.16	X	3.4 - 3.24	-
Theodolite system	4.6.2.3	X	3.4.7	-
Humidity:	4.6.2.20	X	3.4.8	-

TABLE I. Test schedule - continued

Test 1	Paragraph 2	First Article 3	Requirement Paragraph 4	Quality Conformance 5
Telescope:				
Effective aperture	4.6.2.6.1	X	3.5.1.1	-
Field of view	4.6.2.6.2	X	3.5.1.1	-
Minimum focus	4.6.2.6.3	X	3.5.1.1	-
Resolution	4.6.2.6.4	X	3.5.1.1	X
Magnification	4.6.2.6.5	X	3.5.1.1	-
Reticle lines:				
Double	4.6.2.7.1	X	3.5.1.2	-
Single	4.6.2.7.2	X	3.5.1.2	-
Stadia	4.6.2.7.3	X	3.5.1.2	-
Solar circle	4.6.2.7.4	X	3.5.1.2	-
Vertical circle indexing and vertical collimation:				
Altitude level	4.6.2.8.1	X	3.7.3	X
Automatic	4.6.2.8.2	X	3.7.3	X
Optical plummet:	4.6.2.11	X	3.5.3	-
Stadia distance:	4.6.2.12	X	3.5.1.2	-
Tribrach:				
Base plate nut	4.6.2.13.1	X	3.9	-

4.6.2.1 Conditions of test. Prior to beginning the tests, the contractor shall adjust the theodolite by established methods to operate as specified herein. The contractor shall certify that the equipment is ready for the inspections and the testing as specified in table I.

4.6.2.2 Prior to assembly.

4.6.2.2.1 Circles. Horizontal and vertical circles shall be tested for accuracy. In carrying out these tests, the circles shall be mounted on a precise turntable with fixed microscopes located so that a comparison can be made of nominally equal angles on different parts of the circle. Any other method of test giving equal results is acceptable. Evidence of a graduation error exceeding 2 seconds or 0.010 mil, as applicable, for horizontal circles and 3 seconds or 0.015 mil for vertical circles shall constitute failure of the test.

4.6.2.2.2 Level sensitivity. The sensitivity of each level vial shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. The plane of the graduated vials shall be changed a sufficient amount to cause a change of not less than two graduations in the position of the level bubble. The plane of the

ungraduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble as indicated on a scale placed adjacent to the vials. Evidence that the plate level and altitude levels do not have a sensitivity of 20 seconds, ±4 seconds, and that the circular level of the tribrach does not have a sensitivity of 10 minutes, ±3 minutes, shall constitute failure of this test.

4.6.2.2.3 Reticle line weight. The reticle line weights shall be determined by using a measuring microscope or by using any other method that will yield comparable values. Evidence of the reticle line weights differing from 2.5 to 4.0 microns shall constitute failure of this test.

4.6.2.3 Theodolite system (horizontal). The test shall be reported on a test report form similar to that shown in figure 6. A duplicate of the complete test report form shall be placed within the carrying case of the instrument to which the report applies. The test shall be conducted in accordance with the following instructions which shall be printed as part of the test report form:

> This test is based on a series of observations on two collimators approximately 36 degrees (640 mils) apart and on a plane having an elevation approximately the same as that of the horizontal axis of the instrument being tested. In making the following observations, each circle reading shall be estimated to the nearest tenth of a graduation of the breakdown scale. This test measures the following parameters of the theodolite:
>
> a. Ability of the instrument to repeat measurements
> b. Average circle graduation error
> c. Collimation error

Two sets of measurements will be made. Each set will consist of 20 angles (40 measurements) by using all sections of the horizontal circle as follows:

> With the telescope in the direct position, the instrument is pointed on the left collimator with a circle reading of approximately 0 degrees and 36 minutes (5 mils). The circle is read and the seconds (mils) are recorded on line 1, column I. The angle is then turned to the right collimator, the circle read, and the seconds (mils) recorded on line 2, column I. The telescope is then reversed and pointed on the left collimator, the circle read, and the seconds (mils) recorded on line 1, column II (this will be 180 degrees (3200 mils) rather than 0 degrees (0 mils) as with the telescope direct). The angle is then turned to the right collimator, the circle read, and the seconds (mils) recorded on line 2, column II. The mean of the two readings (columns I and II) is entered in column III. Repeat this technique by changing the position of the horizontal circle to read between 36 and 72 degrees (640 - 1280 mils). Advance the circle in this fashion until all parts of the circle are covered during the test.

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The second set of measurements shall be made identically and shall be recorded in columns IV and V. The mean of these readings shall be in column VI. Taking the difference between the two mean readings from column III, angle 1 shall be determined and shall be written in column VII. Like-wise, the difference between the two mean readings from column VI shall be made and angle 2 shall be determined and written in column VIII. The difference of these two angles represents the pointing and reading error of the test (Δ). This shall be written in column IX. Column X is Δ^2.

From column X, $M_2^2 = \frac{\Sigma\Delta^2}{n}$ shall be calculated where n represents the total number of readings. In the case of 36 degrees (640 mils), n = 40, therefore

$$M_2^2 = \frac{\Sigma\Delta^2}{40} \leq 4 \text{ sec}^2 \ (0.0001 \text{ mil}^2).$$

Numbers for M_2^2 larger than 4 sec^2 (0.0001 mil^2) indicate excessive pointing and reading error. The test setup should be checked and, after correction, two new complete sets of readings shall be made. Excessive values of M_2^2 the second time indicated that the instrument is incapable of accurately repeating readings and shall constitute failure of this test.

If M_2^2 is smaller than 4 sec^2 (0.0001 mil^2), proceed with calculations as follows:

Means of the two angles are determined from columns VII and VIII and entered in column XI, mean. These are summed, and a mean is again determined, (M1). Residuals are found and entered in column XII, (V). Column XII shall be scanned. Maximum plus and minus deviations shall be determined and underlined. Any value in excess of 2.0 sec (0.010 mil) shall constitute failure of this test. If plus or minus maximum value is not excessive, compute mean (V). Also compute V^2 and enter in column XIII. Zero is entered opposite the first residual from column XII and the algebraic sum of the residuals is entered in column XIV. These values are added algebraically, and a mean is determined. This mean is subtracted algebraically from the values of column XIV to give column XV (t). The summation of the column should equal zero. Column XV represents the systematic error of the instrument. Column XVI is t^2. Column XVII is the difference between column I and column II (D_1-R_1). The sum of this column divided by twice the amount of readings equals the collimation error. In the case of 36 degrees (640 mils), collimators 12 $C = \frac{\Sigma\ (D_1 - R_1)}{40} \leq 12$ sec (0.060 mil). Any value exceeding sec (0.060 mil) constitutes failure of this test.

The circle graduation error t is computed as follows:

$$M_1^2 = \frac{\Sigma\ V^2}{n_< - 1} = \frac{\Sigma\ V^2}{9}$$

$$M_2^2 = \frac{\Sigma\Delta^2}{n_R} = \frac{\Sigma\Delta^2}{40}$$

$$2t^2 = M_1^2 - M_2^2$$

$$\text{if} \quad M_1^2 - M_2^2 \geq 0$$

$$\text{then} \quad t = \sqrt{\frac{M_1^2 - M_2^2}{2}} \leq 2 \text{ sec } (0.010 \text{ mils})$$

Exceeding this value constitutes failure of this test.

$$\text{if} \quad M_1^2 - M_2^2 < 0$$

$$\text{then compute} \quad (t)m^2 = \frac{\Sigma\ (t)}{n_<} = \frac{\Sigma\ (t)}{10}$$

$$(t)m = \sqrt{\frac{\Sigma\ (t)}{10}}$$

Note: (t) m is not the true circle graduation error; it is the average systematic that includes the circle graduation error.

Since $M_2^2 \leq 4$ sec^2 (0.0001 mil^2) was earlier established, we know that the $t \leq 2$ sec (0.010 mil). This is only true if $M_1^2 - M_2^2 < 0$. The number of angles is designated by $n_<$ and $n_=$ is the number of readings.

A copy of the completed test data sheets or computer printout of same shall be enclosed with each instrument. Absence of data sheets shall constitute failure of this test.

4.6.2.4 Micrometer. The run of the micrometer shall be tested by making coincidence between graduations on opposite sides of the circle with the micrometer reading approximately zero. The exact micrometer reading shall be recorded. The micrometer knob shall then be turned until coincidence is again made near the upper end of the micrometer, and the exact micrometer reading shall be recorded. Three sets of readings in each quadrant of the circle shall be made. The mean reading for each end of the micrometer scale shall be calculated. Subtract the smaller mean from the larger mean. Any difference greater than ±1 second (0.005 mil) for the horizontal circle and 2 seconds (0.01 mil) for the vertical circle shall constitute failure of this test.

4.6.2.5 Collimation.

4.6.2.5.1 Focus-collimation error. Focus-collimation error shall be determined by observing direct and reverse observations on targets at 15 feet (4.6 meters) and infinity (300 feet or 100 meters) and determined as follows:

a. With the telescope in the direct position, sight on the target in the collimator and read the horizontal circle.
b. Transit the telescope to the reverse position and sight on the target in the collimator and read the horizontal circle. Add 180 degrees to this reading.
c. Subtract the smaller reading from the larger reading, and then divide by 2 and this equals the collimation error.
d. The difference between the error at 15 feet and the error at infinity is the focus collimation error.

Evidence of a focus collimation error of more than 10 seconds or 0.05 mil shall constitute failure of this test.

4.6.2.6 Telescope.

4.6.2.6.1 Effective aperture. A scale graduated in millimeters shall be placed in contact with the front of the objective lens. The effective aperture is the diametrical distance of the lens. An aperture of less than 40 mm shall constitute failure of this test.

4.6.2.6.2 Field of view. The field of view shall be measured by pointing the telescope on a collimator or on a target not less than 300 feet (100 meters) from the theodolite. The target shall be brought to the intersection of the edge of the field and the horizontal crossline, and the horizontal circle shall be read. The target shall then be brought to the intersection of the horizontal crossline and the opposite edge of the field, and the circle shall again be read. The difference in circle readings is the field of view. The field of view in the vertical plane shall be measured as specified herein, except that the target shall be brought to the intersections of the vertical crossline and the edges of the field and the readings shall be taken from the vertical circle. Evidence of either field of view measuring less than 1 degree 20 minutes for type I instruments or 24 mils for type II instruments shall constitute failure of this test.

4.6.2.6.3 Minimum focus. The telescope shall be focused on a collimator, the reticle of which has been set to simulate 7 feet (2.2 m), or the telescope shall be set at minimum focus and a graduated scale shall be moved along the line of sight until the markings are most sharply defined. The distance from this point of the scale measured along the optical axis to the vertical axis of the instrument is the minimum focus. Evidence of the minimum focus being more than 7 feet (2.2 m) shall constitute failure of this test.

4.6.2.6.4 Resolution. Observations for resolution shall be made within the stadia lines, employing charts contained in NBS Special Publication 374. Resolution of Pattern 56 of the high contrast chart, with the chart placed 77 feet (23.5 meters) from the telescope, indicates a resolution of 4 seconds, or

better, of arc. Inability of the telescope to resolve the pattern shall constitute failure of this test.

4.6.2.6.5 Magnification. The magnifying power, in diameters, of the telescope shall be measured by focusing the telescope for parallel light. Without change of focus, a transparent graduated scale shall be placed in front of the objective lens and as near thereto as practicable. A measuring microscope shall be focused on the image of the scale. The magnification of the telescope shall then be determined in diameters as the ratio of a given length of the scale to the length of its image in the exit pupil. When an autocollimation eyepiece is furnished, the test shall be performed in the same way. Any other method yielding values accurate within 2 percent may be used in lieu of the specified method. Evidence of a magnification for the telescope differing from 28 diameters, +4 or -1 diameter, or of the autocollimation eyepiece having a magnification less than 18 diameters shall constitute failure of this test.

4.6.2.7 Reticle lines.

4.6.2.7.1 Double vertical reticle lines. With the theodolite sighted on a collimator, the interval between the double vertical lines shall be measured by means of the horizontal circle. Readings differing from an apparent angle of 40 seconds, +5 seconds or -10 seconds, shall constitute failure of this test.

4.6.2.7.2 Single vertical reticle line. With the theodolite sighted on a collimator, the interval between the inner edge of a double vertical reticle line and the adjacent edge of the single vertical reticle line shall be measured. The interval between the inner edge of the other double vertical reticle line and the adjacent edge of the single vertical line shall be measured in the same manner. These angles should be equal. An error of more than ±0.5 second shall constitute failure of this test.

4.6.2.7.3 Stadia lines. With the theodolite sighted on a collimator, the length of the horizontal and vertical stadia lines shall be measured by means of the horizontal and vertical circles. Any measurement less than 5 minutes or more than 7 minutes of arc shall constitute failure of this test.

4.6.2.7.4 Solar circle. With the theodolite sighted and centered on a collimator, the theodolite shall be rotated horizontally 8 minutes with the angle measured by means of the horizontal circle. Next the theodolite shall be rotated vertically until an arc of the solar circle is intersected. The vertical angle (V) is than determined by means of the vertical circle. The solar circle radius (r) can now be calculated $r = 64 + V2$. Calculated measurements differing from 15 minutes 30 seconds ±10 seconds of arc shall constitute failure of this test.

4.6.2.8 Vertical circle indexing and vertical collimation.

4.6.2.8.1 Altitude level. The collimation error of the vertical circle shall be determined by sighting in the direct position on a mark with the horizontal crossline of the reticle, bringing the altitude level to center, obtaining coincidence on the micrometer, and reading the circle. The operation is repeated in the reverse position. The sum of the readings should equal 360 degrees or 6400 mils. Any difference includes twice the vertical collimator error. A net error of more than 12 seconds or 0.060 mil shall constitute failure of this test.

4.6.2.8.2 Automatic. The collimation error of the vertical circle with a self-indexing device shall be determined by sighting in the direct position on a mark with the horizontal crossline of the reticle, obtaining coincidence on the micrometer, and reading the circle. The operation is repeated in the reverse position. The sum of the readings should equal 360 degrees or 6400 mils. Any difference includes twice the vertical collimation error. A net error of more than 12 seconds or 0.060 mil, or failure of the device to damp within 5 seconds of time shall constitute failure of this test.

4.6.2.9 Height of standards. A target placed 30 degrees (530 mils) above the horizon of the instrument and a millimeter scale placed at the same angle below the horizon shall be used to test the standards. The telescope shall be pointed to the target, then tilted downward onto the scale and the reading shall be recorded. The telescope shall be transited and another pointing made. Three sets of direct and reversed readings shall be obtained. The angle formed by the target as apex and the two points where the scale has been read is equal to double the angle of tilt of the trunnion axis with respect to the perpendicular of the vertical axis of the instrument. Evidence of any error more than 12 seconds or 0.06 mil of arc for the sets of readings after correction for the allowable collimation error shall constitute failure of this test.

4.6.2.10 Mirror adjustment. The mirror or mirrors shall be adjusted both vertically and horizontally while the micrometer is observed. Evidence of any mirror adjustment causing more than 1 second or 0.005 mil variation, as applicable, in the circle reading shall constitute failure of this test.

4.6.2.11 Optical plummet.

4.6.2.11.1 Focusing. Failure of the optical system of the plummet to produce a clear image throughout the operating range from tripod height to not less than 110 feet shall constitute failure of this test.

4.6.2.11.2 Integral with alidade. The optical plummet shall be tested for alignment with the vertical axis by viewing a target and rotating the alidade 360 degrees. Evidence that the line of sight of the optical plummet is not concentric with the vertical axis of the theodolite on all points of the circle shall constitute failure of this test.

4.6.2.11.3 On tribrach. The theodolite shall be centered over a ground marker by the plumb bob. The plumb bob shall be removed and the centering

checked by means of the optical plummet. This test shall be repeated with the optical plummet over each side of the tripod head. Evidence that the line of sight is not concentric with the vertical axis shall constitute failure of this test.

4.6.2.12 Stadia distance. With the theodolite leveled on a stand or tripod, a graduated rod set at a known distance over 100 feet (30.5 meters) from the instrument shall be sighted on. The stadia interval shall be read by means of the stadia hairs, first with the rod in the horizontal position and then in the vertical position. This interval shall be multiplied by 100. Evidence of the stadia distance being in error more than ±0.3 percent of the measured distance shall constitute failure of this test.

4.6.2.13 Tribrach.

4.6.2.13.1 Base plate nut. The base plate nut shall be tested for conformance to FED-STD-H28 with certified thread plug gauges. Evidence of any base plate nut not mating with the gauge shall constitute failure of this test.

4.6.2.13.2 Levelment. The theodolite shall be mounted on a tripod or on a trivet which will be tilted 6 degrees from the horizontal. The instrument shall be leveled by means of the tribrach foot screws. Any evidence that the instrument cannot be leveled shall constitute failure of this test.

4.6.2.14 Rainfall.

4.6.2.14.1 Theodolite. The theodolite, with the eyepiece down, shall be placed in a rain-testing facility not less than 36 inches high, 36 inches wide, and 36 inches deep (1 meter^3). The facility shall be equipped with water spray nozzles to provide a uniform flow of water over the test area at a rate of 3 inches (8 cm), ±1 inch (2 cm), per hour. The direction of rainfall shall be variable up to 45 degrees from the vertical. Water temperature shall be 52 ^{0}F (11 ^{0}C) to 68 ^{0}F (20 ^{0}C). The theodolite shall be within ±5 ^{0}F of the water temperature. The theodolite shall be subjected to this test for 5 minutes. Presence of water in the telescope or optical system or bearings shall constitute failure of this test.

4.6.2.14.2 Rainfall test of carrying case. The theodolite in the carrying case shall be subjected to the test specified in 4.6.2.14.1 for a period of 1 hour. Presence of water in the carrying case, the telescope, or optical system shall constitute failure of this test.

4.6.2.14.3 Rainfall test of transport case. The rainfall test of the transport case may be conducted concurrently with the rainfall test of the carrying case, except that the transport case shall be subjected to the rainfall test specified in 4.6.2.14.1 for a period of 1 hour. Presence of water in the transport case, swelling or buckling, loose hardware, or inability to close the case shall constitute failure of this test.

4.6.2.15 Sand and dust.

4.6.2.15.1 Theodolite. The theodolite shall be placed in a sand and dust testing chamber not less than 36 inches high, 36 inches wide, and 36 inches deep (1 meter^3). The sand and dust (140-mesh silica flour (see 6.4)) shall pass through the chamber at a velocity of 100 to 500 feet per minute (30 to 152 meters). The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature shall be maintained at 77 °F (25 °C), ±5 °F (±3 °C). The relative humidity shall not exceed 30 percent. The theodolite shall be subjected to this test for a period of 15 minutes. The presence of sand or dust in the telescope, in the optical system, or on bearings, or the presence of lens abrasion shall constitute failure of this test.

4.6.2.15.2 Carrying case. The carrying case shall be subjected to the test specified in 4.6.2.15.1 for a period of 1 hour. Presence of sand or dust in the carrying case shall constitute failure of this test.

4.6.2.16 Operational. The operational test shall be conducted at each time indicated in table I. The theodolite shall be operated and checked as follows to determine its operational capabilities:

a. Manipulate all movable parts and check for freedom of movement and noninterference with other parts in various operating positions throughout their maximum ranges.
b. Fit or attach all accessories to the theodolite and check for functionality as specified.
c. Check the levels for free movement to determine if the bubble remains within the graduations and if the bubble ends of the coincidence level remain visible.
d. Rotate the theodolite about its horizontal and vertical axes, and check the illumination system for uniform and steady operation.
e. Check microscope focus for uniform and clear focus around each entire circle.
f. Check parts, components, accessories for damage.

Evidence that the theodolite components, or accessories will not function as specified, presence of broken, cracked, or shifted parts on the theodolite or accessories, or opened joints, loose hardware, or splintered, cracked, or swollen wood on the cases shall constitute failure of this test.

4.6.2.17 Low temperature and high temperature.

4.6.2.17.1 Low temperature and high temperature test chamber. The low temperature tests shall be conducted in a test chamber of a size that will house the theodolite mounted on its tripod, the cases and accessories, and shall permit the inspector to walk around the periphery of the tripod legs in making the examination and tests specified herein.

4.6.2.17.2 Low temperature operation. The theodolite shall be covered with the waterproof cover and the theodolite, carrying case, transport case, battery case (batteries excluded), and accessories shall be placed in a temperature chamber. The temperature in the chamber shall be gradually lowered to -40 °F (-40 °C) and maintained at that temperature until the theodolite is in thermal equilibrium. The batteries (which are not subject to this test) shall then be placed in the battery box and the theodolite shall be connected to the battery box. The theodolite shall then be subjected to the operational test specified in 4.6.2.16. The waterproof cover shall be folded four times. Loss of flexibility or cracking shall constitute failure for the cover. Inability of the instrument, with winterization kit installed, and of the accessories to operate at -40 °F (-40 °C) shall constitute failure of this test.

4.6.2.17.3 Low temperature storage. Upon completion of the operational test at low temperature, the theodolite shall be placed in its carrying case which is then placed in the transport case. The tripod shall be folded and all accessories stowed for transport. The temperature of the test chamber shall be lowered to -50 °F (-45.6 °C) and maintained at this temperature for 6 hours. The temperature shall then be gradually raised to that of the laboratory and the theodolite dried out and subjected to the operational test specified in 4.6.2.16. Evidence of damage to the theodolite or accessories during storage, or inability of the theodolite to complete the operational test shall constitute failure of this test.

4.6.2.17.4 High temperature operation. The theodolite, mounted on the tripod, carrying case, transport case, battery case (batteries included), accessories, and waterproof cover shall be placed in a temperature control chamber. The temperature in the chamber shall be gradually raised to 125 °F (51.7 °C) and remain at this temperature until the theodolite is in thermal equilibrium. Relative humidity shall be maintained between 14 and 44 percent. The theodolite, cases, and accessories shall then be tested as specified in 4.6.2.16. Evidence of damage to the waterproof cover and inability of the theodolite and accessories to operate shall constitute failure of this test.

4.6.2.17.5 High temperature storage. Upon completion of the high temperature operational test, the theodolite shall be placed in its carrying case which is then placed in the transport case. The tripod shall be folded and all accessories stowed for transport. The temperature of the test chamber shall then be raised to 155 °F (68.3 °C) and the temperature maintained for 4 hours. Relative humidity shall be maintained between 14 and 44 percent. The temperature shall then be gradually lowered to that of the laboratory and the theodolite shall again be subjected to the operational tests of 4.6.2.16. Evidence of damage to the theodolite and accessories during storage shall constitute failure of this test.

4.6.2.17.6 Solar radiation. The theodolite shall be temperature stabilized at 125 °F (51.7 °C) and a source of directed radiant energy simulating solar radiation of 355 Btu/ft^2/hr shall be applied to the theodolite for a period of 4 hours. At the end of this period, the theodolite shall be tested as

specified in 4.6.2.16. In addition, the theodolite shall be observed for peeled, cracked, or blistered paint. Presence of any of the defects specified in 4.6.2.16 or of peeled, cracked, or blistered paint shall constitute failure of this test.

4.6.2.17.7 Temperature test of the wiring. The temperature test of the wiring shall be conducted by using the battery and hand lamp cords. Temperature tests of the wiring may be conducted concurrently with the temperature tests of the theodolite.

4.6.2.17.7.1 Low temperature operation test of wiring. The battery and hand lamp cords shall be in the battery and lighting accessory case when placed in the temperature control chamber. The two cords shall remain in the control chamber during the low temperature operation test of the theodolite. At the end of the test, and without being removed from the chamber, the two cords shall be bent around a 3/8-inch (9.5 mm) mandrel at a rate of 12 turns per minute. The two cords shall then be removed from the chamber and examined for cracks or flaws under a glass having a magnification of 3 diameters. Evidence of damage shall constitute failure of the test.

4.6.2.18 Vibration. The theodolite shall be subjected to vibration testing in accordance with the following.

4.6.2.18.1 Carrying case removed from the transport case. The theodolite in the carrying case shall be rigidly mounted to the vibration table and subjected to two sweeps of sinusoidal cycling vibration in each of three mutually perpendicular axes. Each sweep shall be 5 to 55 to 5 Hz and, traversed in 15 minutes. The vibration amplitude shall be 0.10 inch double amplitude from 5 to 17 Hz and 1.5 g from 17 to 55 Hz. Upon completion of this test, the theodolite shall be mounted on its tripod and tested for operation as specified in 4.6.2.16. Evidence of damage to the theodolite or the case shall constitute failure of the test.

4.6.2.18.2 Transport case. The theodolite in the carrying-transport case shall be tested in accordance with MIL-STD-810, procedure X, table 514.2-VII, figure 514.2-7, curve AX with the exception that only five sweeps of vibration cycling shall be performed for a total of 1 hour of vibration in each axis. Upon completion of this test, the theodolite shall be mounted on its tripod and tested for operation as specified in 4.6.2.16. Evidence of damage to the theodolite or the case shall constitute failure of this test.

4.6.2.19 Shock. The theodolite in the carrying case shall be flat dropped in an upright position from a height of 6 inches (15.3 cm) onto a level wood floor. The theodolite shall then be subjected to the operational test specified in 4.6.2.16. The theodolite in its carrying case shall then be placed in the transport case and, after being secured for shipment, shall be dropped from a height of 12 inches (30 cm) so that one bottom edge of the transport case strikes a level concrete floor when the bottom of the transport case is inclined 15 to 20 degrees from the horizontal. The theodolite shall

again be tested as specified in 4.6.2.16. Evidence of damage to the theodolite or to the cases shall constitute failure of this test.

4.6.2.19.1 Shock test of transport case. A dummy load in the configuration of the carrying case, and equal in weight to the theodolite and the carrying case, shall be placed in the transport case. The transport case shall be closed and secured as for shipment and subjected to eight drops, one on each bottom edge and one on each corner onto horizontal concrete floor from a height of 42 inches (1.06 meters). The height of drop shall be measured from the concrete floor to the nearest edge or corner of the transport case when suspended prior to each drop. The presence of splinters or cracks, broken, cracked or loose hardware, or opened joints, shifted padding or deformation of the transport case shall constitute failure of this test.

4.6.2.20 Humidity. Place the theodolite with power off, mounted in the operating position, with all accessories, in a humidity chamber conforming to MIL-STD-810, method 507.2, procedure II. The theodolite shall be inspected and tested as specified in 4.6.2.16. Evidence of damage to the theodolite or to the cases shall constitute failure of this test.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete theodolite with all components and accessories provided, shall be preserved, packed and marked in accordance with the level A, B, or C requirements of MIL-S-3534 as specified (see 6.2).

6. NOTES:

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Type I theodolites are primarily intended for second-order survey and azimuth work. Type II theodolites are primarily intended for artillery and missile survey.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of theodolite required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of the first article (see 3.2).
e. When the Government will conduct any or all of the first article inspection and tests. When the Government will conduct some but not

all of the first article inspection and tests, the contracting officer should specify which inspection and tests will be conducted by the Government and which inspection and tests shall be conducted by the contractor (see 3.2).

f. When the reticle shall be other than as specified (see 3.5.1.2).
g. When an elbow eyepiece for the telescope is not required (see 3.5.1.4).
h. When a solar filter is not required (see 3.5.1.5).
i. When an autocollimation eyepiece is required (see 3.5.1.6).
j. When the sectors shall be frosted (see 3.5.3).
k. When a circle selector device is required (see 3.6.5 and 6.7).
l. When an elbow eyepiece for the microscope is not required (see 3.6.6).
m. When an altitude level is required (see 3.7.3).
n. When a hand lamp is required (see 3.12.2.2).
o. When a battery and lighting accessory case is required (see 3.12.2.4).
p. When composition II is used (see 3.12.2.5.1).
q. When a knapsack is required (see 3.17).
r. Color required (see 3.17 and 3.20).
s. When tools and accessories are required (see 3.21).
t. When a tripod is required (see 3.22.3 and 3.23.6).
u. When optical plummet shall be illuminated (see 3.23.3).
v. When a clamp assembly is required (see 3.23.4).
w. When a collimation light assembly is required (see 3.23.5).
x. Level of preservation, packing and marking required (see 5.1).

6.3 First article. When a first article inspection is required the item(s) should be a preproduction model. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitations for bid should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Sand and dust. The sand and dust to be used in the sand and dust test is commercially known as "140-mesh silica flour". Sand and dust, 140-mesh flour produced by the Fenton Foundry Supply Company, Dayton, OH 45403, and Ottawa Silica Company, Ottawa, IL 61350, or equal, has been found satisfactory.

6.5 Information figures. Figure 1 shows a type of stub which has been found acceptable and figure 2 shows a reticle pattern which has been found acceptable; however, these figures have been included for illustration only and are not intended to preclude the furnishing of other means of attaching the alidade to the tribrach or of other reticle patterns which conform to this specification.

Custodians: Army - MI
Navy - OS
Air Force - 17
DLA - CC
Review activities:
Army - AR, AT, AV, CR, EA, GL, MD, SM
Navy - AS, CG, EC, MC, SH, TD
Air Force - 11, 19, 99
DLA - DH, DP, DS
User activities:
Army - AA, AC, AL, AM, AS, AT, AV, CE, CR, EA, EL, GL, GS, HA, MD, MI, MP, SM, TE, TM
Navy - AS, CG, EC, MC, OS, SA, SH, TD
Air Force - 03, 04, 05, 11, 13, 19, 20, 23, 26, 27, 29, 30, 37, 38, 41, 42, 43, 44, 45, 46, 50, 52, 55, 57, 60, 61, 62, 63, 65, 68, 70, 71, 72, 73, 74, 75, 76, 77, 79, 80, 81, 82, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99
DLA - DH, DP, DS
Preparing activity:
Army - MI
(Project 6675-0001)

6.6 Second-order accuracy. Information regarding second-order accuracy will be found in Department of the Army Technical Manual TM 5-441, Topographic Surveying.

6.7 Circle selector. The circle selector is used for blanking out either the horizontal circle or the vertical circle so that only one circle can be seen at one time. The circle selector should be specified when the theodolite is to be used with missiles.

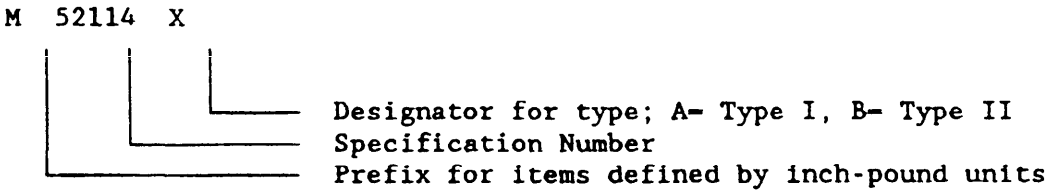
6.8 Technical Manual. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be listed on a separate Contract Data Requirements List (DD form 1423), which is included as an exhibit to the contract. The Technical Manual must be acquired under separate contract line item in the contract.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of theodolites.

6.10 Subject area term (key word) listing.

a. Alidade
b. Azimuth work
c. Missile survey
d. Optical reading

6.11 Part or identifying number (PIN). The PIN to be used for theodolites acquired to this specification are created as follows:



6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activities:
DLA - GS

User activity:
Navy - MC
AF - 82

Preparing activity:
Army - ME

Project 6675-0936

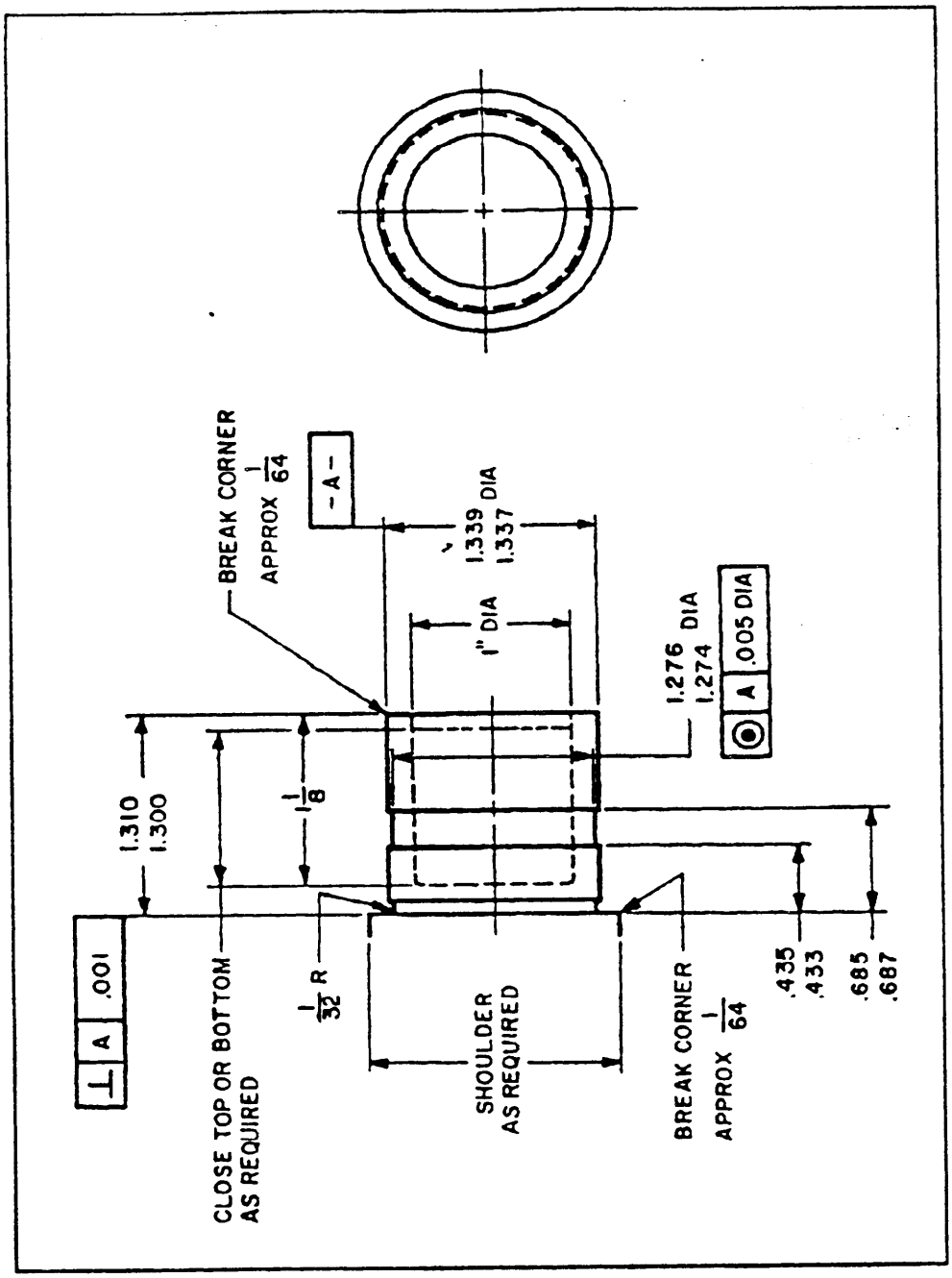


FIGURE 1. Theodolite Stub

CX-1563

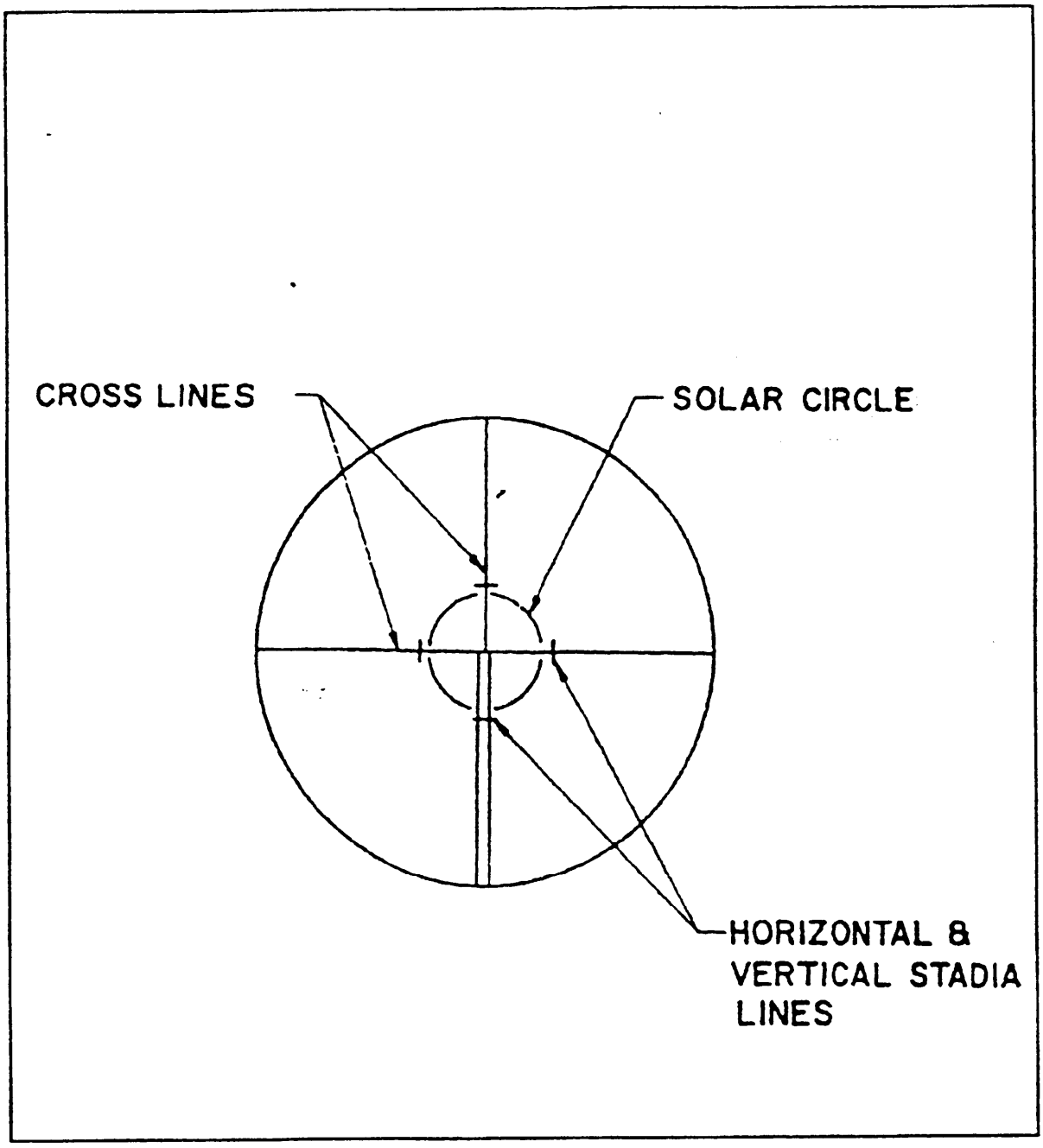
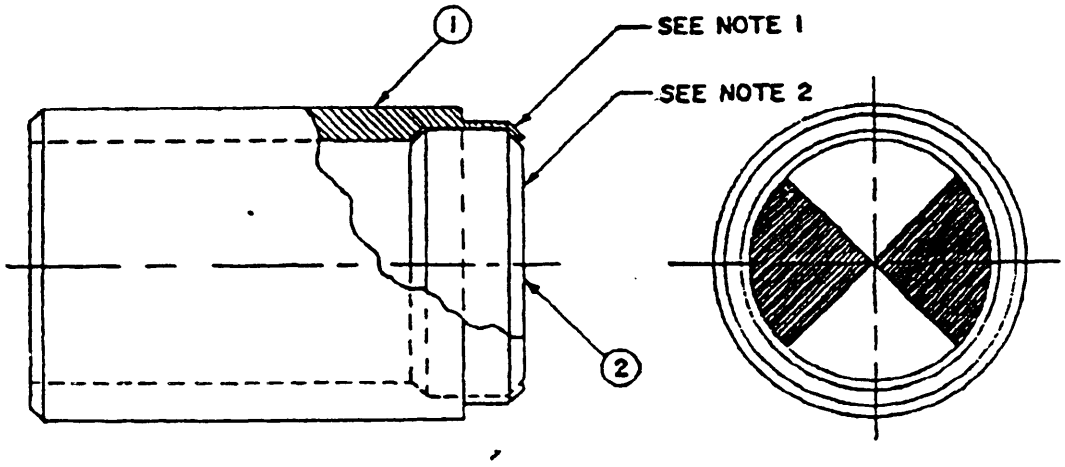


FIGURE 2. Reticle Pattern for Theodolite

X-3104A

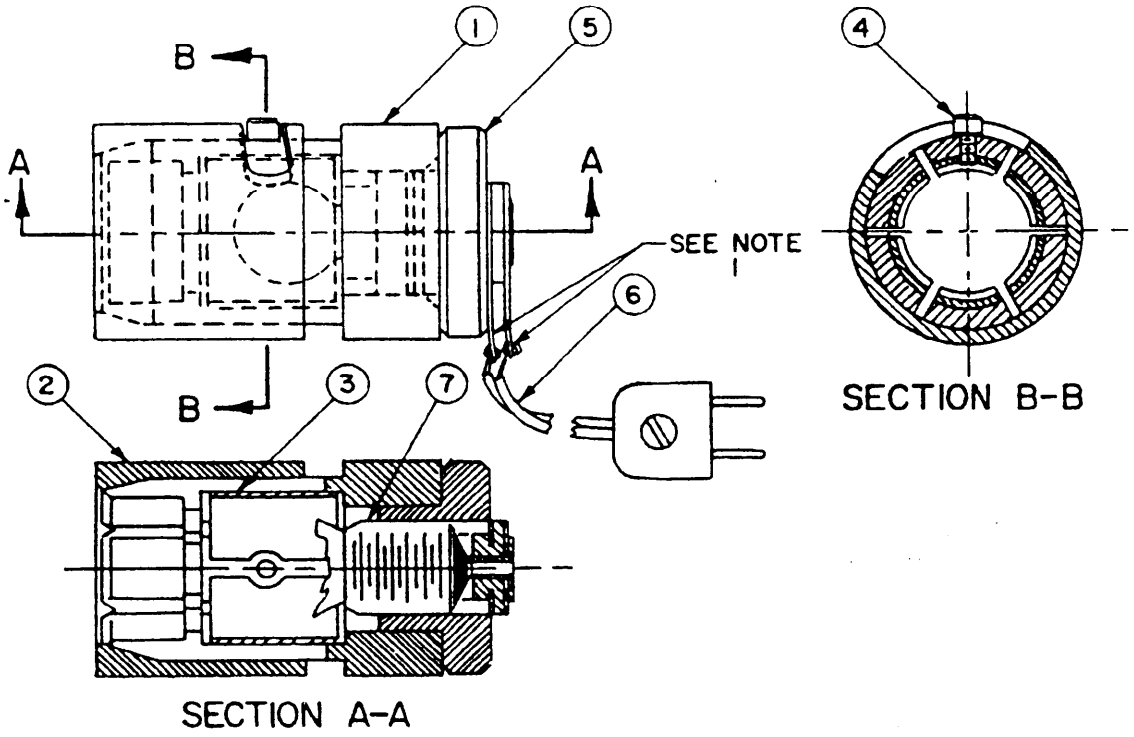


QT'Y REQ'D	FIND NO.	PART NAME	ORD. PART NO.
1	1	MOUNT, RETICLE	10018573
1	2	RETICLE MODIFICATION	10101815

NOTES:
1. ROLL EDGE OF MOUNT TO SECURE RETICLE.
2. INSTALL WITH RETICLE HAIRLINE OUTSIDE OF RETICLE MOUNT.
3. APPLY PART NUMBER PER MIL-STD-130 BY TAG, 1/8 HIGH CHARACTERS.

FIGURE 3. **Reticle Modification Assembly**

CX1729



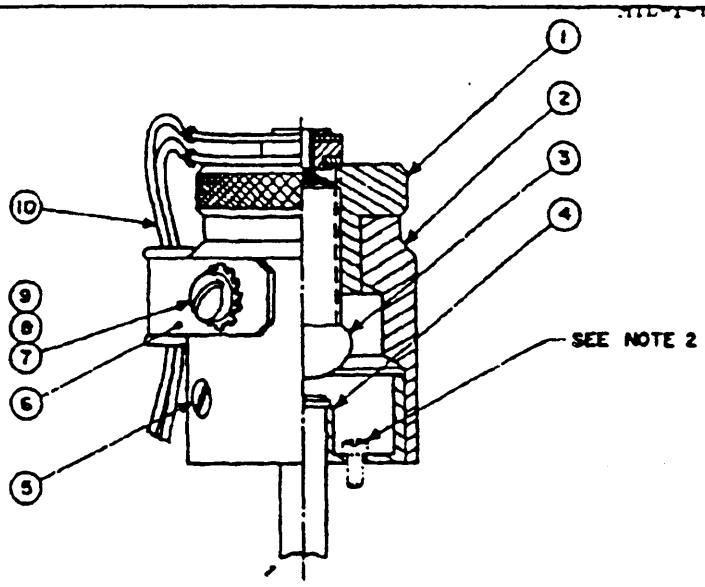
QT'Y REQ'D	FIND NO.	PART NAME	ORD. PART NO.
1	1	CONNECTOR	10102204
1	2	SLEEVE	10102203
1	3	RING	10102205
1	4	SCREW, CAP	10102206
1	5	SOCKET ASSEMBLY	10010896
1	6	CABLE ASS'Y S.P.	10194010
1	7	LAMP, INCANDESCENT	MS15611-5

NOTES:

1 SOLDER WITH QQ-S-571 (SN60) PER DOD-STD-1866 USING ROSIN FLUX. SOLDERED JOINTS AND ADJACENT WIRE SHALL BE COATED WITH MPD 9196 MATERIAL.

2 APPLY PART NUMBER PER MIL-STD-130 BY TAG, 1/8 HIGH CHARACTERS.

Figure 4. EYE-PIECE ADAPTER ASSEMBLY.



QTY REQD	FIND NO.	PART NAME	ORD PART NO.
1	1	SOCKET ASSY	10010896
1	2	ADAPTER	10061272
1	3	LAMP, INCANDESCENT	10009795
1	4	COLLIMATION TARGET LENS ASSY	10061269
1	5	SCREW, FLAT HEAD	MS 51959-2
1	6	CLAMP, LOOP	10003908
1	7	SCREW, PAN HEAD	MS 51957-13
1	8	WASHER, LOCK, FLAT, EXT TOOTH	MS 35335-29
1	9	WASHER	9126409
1	10	CABLE	10011833

NOTES:

1. APPLY PART NUMBER PER MIL-STD-130 BY TAG, $\frac{1}{8}$ HIGH CHARACTERS.
2. FASTEN COLLIMATION TARGET ASSEMBLY TO TELESCOPE WITH EXISTING THEODOLITE SCREWS AND APPLY ELECTRICAL INSULATION COATING PER MPD 9044 (2 PLACES).

FIGURE 5. Collimation Light Assembly

DX-1732B

Column		I	II		III	IV	V		VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV	XVI	XVII
Pointing		Mils Min θ_1 Sec	Mils Min θ_2 Sec		$\frac{\theta_1+\theta_2}{2}$	Mils Min θ_3 Sec	Mils Min θ_4 Sec		$\frac{\theta_3+\theta_4}{2}$	<1 (B-A)	<2 (D-C)	$(<1 - <2)$	$\frac{1}{2}$	$\frac{<1 + <2}{2}$	V $(XI - M_1)$	V^2	R $(V_1+V_2+V_3)$	(ϵ) $(2IV - R)$	$(\epsilon)^2$	$\theta_1-\theta_2$
Deg	Mil																			
0	0			A				C												
36	640			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
36	640			A				C												
72	1280			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
72	1280			A				C												
108	1920			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
108	1920			A				C												
144	2560			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
144	2560			A				C												
180	3200			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
180	3200			A				C												
216	3840			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
216	3840			A				C												
252	4480			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
252	4480			A				C												
288	5120			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
288	5120			A				C												
324	5720			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
324	5720			A				C												
0	0			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
														$\Sigma N =$	$\Sigma V_3 =$		$\Sigma R =$		$\Sigma C =$	
												$M_1^2 = \frac{\Sigma^2}{96} =$		$M_2 = \frac{\Sigma}{16} =$	$M_3^2 = \frac{\Sigma^2}{8} =$		$M_3 = \frac{\Sigma}{16} =$		$C = \frac{\Sigma}{48} =$	

$2c^2 = M_1^2 - M_3^2$ If $M_1^2 - M_3^2 \geq 0$ then $c = \pm\sqrt{\frac{M_1^2 - M_3^2}{2}}$

Engr _____ Date _____

Ser. No. _____

Order _____

Tested by _____

Checked by _____

FIGURE 6. Data Sheet for Theodolite System Test (horizontal). X-3034

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-14132G	91/05/07

3. DOCUMENT TITLE
Theodolites, Surveying, Direction: Second Order

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If Applicable)
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	